

NORTHERN AROOSTOOK, ME, USA

WMO: 726083

Lat: 47.286N

Lon: 68.313W

Elev: 988

StdP: 14.18

Time Zone: -5.00 (NAE)

Period: 95-19

WBAN: 04836

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.0	-8.6	-20.6	1.8	-11.6	-17.0	2.2	-7.2	28.7	16.2	26.6	11.4	10.1	320	0.655

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.3	82.4	67.2	79.3	65.2	76.1	63.9	70.0	77.8	68.2	75.5	66.5	73.3	9.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.2	103.9	73.4	65.7	98.3	72.1	63.9	92.2	69.4	34.6	77.7	33.1	75.5	31.7	73.0	78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.9	21.8	19.3		-18.6	87.6	5.0	3.0	-22.1	89.7	-25.0	91.5	-27.8	93.1	-31.4	95.3
				-18.8	73.2	4.7	1.9	-22.1	74.6	-24.9	75.6	-27.5	76.7	-30.9	78.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	39.2	10.3	13.7	23.4	36.6	50.7	59.3	65.0	63.4	55.2	43.0	30.6	17.9	(6)
(7)		DBStd	20.85	11.87	10.94	10.39	7.75	7.78	6.78	5.44	5.71	7.40	7.64	8.88	10.40	(7)
(8)		HDD50	5420	1229	1017	827	405	86	6	0	0	29	243	584	994	(8)
(9)		HDD65	9563	1694	1437	1291	851	447	196	67	101	306	681	1033	1459	(9)
(10)		CDD50	1491	0	0	1	4	108	286	465	414	185	27	1	0	(10)
(11)		CDD65	159	0	0	0	0	4	26	67	50	12	0	0	0	(11)
(12)		CDH74	1155	0	0	0	1	82	218	432	332	87	3	0	0	(12)
(13)		CDH80	226	0	0	0	0	15	43	86	67	15	0	0	0	(13)
(14)	Wind	WSAvg	9.3	10.3	10.4	10.8	10.4	9.4	8.4	7.4	8.3	9.4	9.8	10.0	(14)	
(15)	Precipitation	PrecAvg	38.1	2.2	2.2	2.4	2.6	3.0	4.2	4.3	3.7	3.6	3.5	3.2	3.2	(15)
(16)		PrecMax	49.0	4.6	4.3	4.4	5.2	5.4	6.6	5.9	7.5	7.6	6.7	6.7	5.4	(16)
(17)		PrecMin	29.0	1.0	0.9	0.7	1.1	1.5	2.4	1.3	1.2	1.1	1.6	0.5	1.8	(17)
(18)		PrecStd	5.4	0.9	0.8	1.0	1.1	1.0	1.2	1.2	1.7	1.6	1.4	1.4	1.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.0	42.2	53.6	66.5	81.6	84.6	87.1	85.9	81.9	70.4	59.0	45.2	(19)
(20)			MCWB	42.0	39.6	46.4	50.4	60.5	67.2	68.9	69.3	68.4	58.9	56.0	43.8	(20)
(21)		2%	DB	37.1	36.6	45.1	58.6	75.1	80.0	82.1	81.5	75.6	63.2	52.1	38.7	(21)
(22)			MCWB	35.6	33.7	37.0	46.1	57.6	64.6	67.9	66.9	65.1	54.9	48.7	37.1	(22)
(23)		5%	DB	32.6	33.6	40.6	53.6	70.1	75.5	78.9	77.5	71.9	58.7	46.8	35.2	(23)
(24)			MCWB	31.1	31.4	34.9	43.3	55.3	62.6	66.0	65.2	63.0	53.1	43.8	33.8	(24)
(25)		10%	DB	27.8	30.0	37.2	48.5	64.7	72.1	75.4	74.1	67.9	54.8	43.2	32.3	(25)
(26)			MCWB	26.2	28.1	33.0	40.8	52.5	60.7	64.7	63.5	60.4	49.9	40.3	31.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.4	40.2	47.7	54.6	65.5	71.0	73.3	71.4	70.1	61.1	56.6	44.0	(27)
(28)			MCD B	42.8	41.5	51.4	62.6	76.6	81.3	80.3	81.0	79.2	65.6	59.1	45.0	(28)
(29)		2%	WB	36.0	34.5	38.9	48.6	60.8	67.7	70.3	69.2	67.0	57.3	49.4	37.4	(29)
(30)			MCD B	36.9	36.0	42.2	54.9	69.4	75.3	77.3	76.8	73.2	61.5	51.0	38.1	(30)
(31)		5%	WB	31.0	31.4	36.1	45.0	57.4	65.4	68.4	67.5	64.4	53.8	44.3	33.8	(31)
(32)			MCD B	32.5	32.9	39.8	51.5	66.0	71.9	75.5	74.5	69.5	57.8	46.1	34.8	(32)
(33)		10%	WB	26.4	28.1	33.3	41.7	54.5	63.0	66.6	65.8	61.7	50.6	40.5	31.0	(33)
(34)			MCD B	27.9	29.7	37.1	48.2	63.3	68.5	73.5	71.8	66.6	54.4	43.3	32.1	(34)
(35)	Mean Daily Temperature Range	MDBR	14.7	15.3	15.4	15.2	18.6	18.1	17.3	17.8	17.6	14.0	11.1	12.1	(35)	
(36)		5% DB	MCD BR	18.6	15.0	17.7	24.1	27.5	23.7	20.8	21.6	21.7	20.5	16.3	14.6	(36)
(37)			MCWBR	18.7	13.2	12.9	15.2	14.1	12.2	10.1	10.3	12.5	13.9	14.3	14.3	(37)
(38)		5% WB	MCD BR	18.7	14.7	15.6	20.7	22.6	19.6	17.7	18.3	19.1	17.9	15.5	14.0	(38)
(39)			MCWBR	18.9	14.1	13.0	15.2	13.6	11.7	10.3	10.0	12.9	14.1	14.8	14.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.250	0.263	0.297	0.341	0.345	0.365	0.395	0.373	0.350	0.316	0.286	0.262	(40)	
(41)		taud	2.344	2.299	2.271	2.360	2.439	2.381	2.314	2.379	2.441	2.532	2.528	2.395	(41)	
(42)		Ebn at Noon	270	290	294	289	292	285	274	276	273	268	257	251	(42)	
(43)		Edh at Noon	22	29	36	36	35	37	39	35	30	23	19	19	(43)	
(44)	All-Sky Solar Radiation	RadAvg	432	716	1101	1416	1594	1728	1724	1508	1177	693	386	296	(44)	
(45)		RadStd	54	85	79	118	165	115	161	97	87	63	45	28	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page